Raspberry Pi BME Sensors Tutorial

Contents

Raspberry Pi BME Sensors Tutorial 1
BME280 and BME680 Sensor Documentation 1
BME280 and BME680 1
RealVNC Viewer 2
Tutorial 0: Install BME280 and BME680 Libraries 2
Tutorial 1: Hello BME280! 2
Tutorial 1: Hello BME680! 5
Tutorial 2: BME280 Simple 2 7
Tutorial 2: BME680 Simple 2 9
Tutorial 3: BME680 Simple 3 11
Air Quality 13
bme68x_test.py 13

BME280 and BME680 Sensor Documentation

Pimoroni BME280 sensor documentation: https://github.com/pimoroni/bme280-python

Pimoroni BME680 sensor documentation: https://github.com/pimoroni/bme680-python

BME280 and BME680

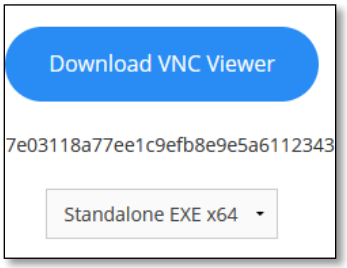
This tutorial is for the Grove - Temperature, Humidity and Pressure BME280 sensor (BME280) and the Grove - Temperature, Humidity, Pressure and Gas Sensor (BME680).

https://www.seeedstudio.com/Grove-BME280-Environmental-Sensor-Temperature-Humidity-Barometer.html

https://www.seeedstudio.com/Grove-Temperature-Humidity-Pressure-and-Gas-Sensor-BME68-p-3109.html

RealVNC Viewer

RealVNC viewer allows us to remotely control the Raspberry Pi in headless mode.



1. Go to https://www.realvnc.com/en/connect/download/viewer/
2. Download the VNC Viewer Standalone EXE anywhere you want to run the program from. You don't have to install it.
3. Double Click **VNC Viewer**.
4. Type in the IP address of your robot → Click **Connect**.

Tutorial 0: Install BME280 and BME680 Libraries

1. Shutdown the Pi. (Do not connect sensors when the Pi has power.)
2. Plug the BME280 or BME680 sensor into an I2C port.
3. Mount the sensor on a sensor mount.
4. Powerup the Pi.
5. Open a terminal.
6. Install BME280 drivers – Docs: https://pypi.org/project/pimoroni-bme280
    a. **sudo pip3 install pimoroni-bme280**
7. Install BME680 drivers - Docs: https://pypi.org/project/bme680
    a. sudo pip3 install bme680

Tutorial 1: Hello BME280!

1. Type: **geany bme280_simple_1.py**
2. This will open the geany code editor and create a Python file.
3. Copy and paste the following code. This code will not change. It imports the sensor library and initializes the sensor.

```
#!/usr/bin/env python3
"""
    Name: bme280_simple_1.py
    Purpose: Use Pimoroni library to read
    temperature, pressure, and humidity from Bosch bme280 sensor
    !Connect to I2C bus
    Press Ctrl+C to exit
"""
from time import sleep
try:
    from smbus2 import SMBus
except ImportError:
    from smbus import SMBus

# sudo pip3 install pimoroni-bme280
from bme280 import BME280

# Initialize the BME280 sensor
bus = SMBus(1)
sensor = BME280(i2c_dev=bus)
```

4. Enter the following code after the copied and pasted section.

```python
#!/usr/bin/env python3
"""
    Name: bme280_simple_1.py
    Purpose: Use Pimoroni library to read
    temperature, pressure, and humidity from Bosch bme280 sensor
    !Connect to I2C bus
    Press Ctrl+C to exit
"""
from time import sleep
try:
    from smbus2 import SMBus
except ImportError:
    from smbus import SMBus

# sudo pip3 install pimoroni-bme280
from bme280 import BME280

# Initialize the BME280 sensor
bus = SMBus(1)
sensor = BME280(i2c_dev=bus)

print("BME280 Read temperature, pressure, and humidity")
print("Ctrl+C to exit!")

while True:
    # Temperature in celsius
    print(sensor.get_temperature())

    # Barometric ressure in hPa (hectopascal)
    print(sensor.get_pressure())

    # Relative humidity in %
    print(sensor.get_humidity())

    sleep(1)
```

Example run:

```
pi@raspberrypi3:~/Code $ python3 bme280_simple_1.py
BME280 Read temperature, pressure, and humidity
Ctrl+C to exit!
23.537171707028754
618.5009584899727
65.11997946527883
```

Tutorial 1: Hello BME680!

1. Type: **geany bme680_simple_1.py**
2. This will open the geany code editor and create a Python file.
3. Copy and paste the following code. This code will not change. It imports the sensor library and initializes the sensor.

```
#!/usr/bin/env python3
"""
    Filename: bme680_simple_1.py
    Purpose: Use Pimoroni library to read
    temperature, pressure, and humidity from Bosch bme280 sensor
    !Connect to I2C bus
    Press Ctrl+C to exit
"""
from time import sleep

# sudo pip3 install bme680
import bme680

# Initialize sensor object, make connection to sensor over I2C
sensor = bme680.BME680(bme680.I2C_ADDR_PRIMARY)
```

4. Enter the following code starting after the copied and pasted section.

```python
#!/usr/bin/env python3
"""
    Filename: bme680_simple_1.py
    Purpose: Use Pimoroni library to read
    temperature, pressure, and humidity from Bosch bme280 sensor
    !Connect to I2C bus
    Press Ctrl+C to exit
"""
from time import sleep

# sudo pip3 install bme680
import bme680

# Initialize sensor object, make connection to sensor over I2C
sensor = bme680.BME680(bme680.I2C_ADDR_PRIMARY)

print("BME680 Read temperature, pressure, and humidity")
print("Ctrl+C to exit!")

while True:
    # Read the sensor
    sensor.get_sensor_data()

    # Temperature in celsius
    print(sensor.data.temperature)

    # Barometric ressure in hPa (hectopascal)
    print(sensor.data.pressure)

    # Relative humidity in %
    print(sensor.data.humidity)

    sleep(1)
```

Example run:

```
pi@raspberrypi3:~/Code $ python3 bme680_simple_1.py
BME680 Read temperature, pressure, and humidity
Ctrl+C to exit!
22.28
890.84
42.284
22.28
890.84
42.284
22.28
```

Tutorial 2: BME280 Simple 2

1. Shutdown the Pi. (Do not connect sensors when the Pi has power.)
2. Plug the BME280 sensor into an I2C port.
3. Mount the sensor on a sensor mount.
4. Powerup the Pi.
5. Open a terminal.
6. Type: **geany bme280_simple_2.py**
7. This will open the geany code editor and create a Python file.
8. Enter the following code.

```python
#!/usr/bin/env python3
"""
    Name: bme280_simple_2.py
    Purpose: Use Pimoroni library to read
    temperature, pressure, and humidity from Bosch bme280 sensor
    !Connect to I2C bus
    Press Ctrl+C to exit
"""
from time import sleep
try:
    from smbus2 import SMBus
except ImportError:
    from smbus import SMBus

# sudo pip3 install pimoroni-bme280
from bme280 import BME280

# Initialize the BME280 sensor
bus = SMBus(1)
sensor = BME280(i2c_dev=bus)

print("BME280 Read temperature, pressure, and humidity")
print("Ctrl+C to exit!")

try:
    while True:
        # Temperature in celsius
        temp_c = sensor.get_temperature()

        # Relative humidity in %
        humidity = sensor.get_humidity()

        # Barometric pressure in hPa (hectopascal)
        pressure_hpa = sensor.get_pressure()

        print(f"{temp_c:.1f} C | {humidity:.0f} % | {pressure_hpa:.2f} hPa")

        sleep(1)

except KeyboardInterrupt:
    print("Bye!")
    exit(0)
```

Example run:

```
pi@raspberrypi3:~/Code $ python3 bme280_simple_2.py
BME280 Read temperature, pressure, and humidity
Ctrl+C to exit!
23.5 C | 65 % | 618.50 hPa
21.8 C | 37 % | 891.07 hPa
21.8 C | 37 % | 891.07 hPa
21.8 C | 37 % | 891.07 hPa
```

Tutorial 2: BME680 Simple 2

1. Shutdown the Pi. (Do not connect sensors when the Pi has power.)
2. Plug the BME680 sensor into an I2C port.
3. Mount the sensor on a sensor mount.
4. Powerup the Pi.
5. Open a terminal.
6. Type: **geany bme680_simple_2.py**
7. This will open the geany code editor and create a Python file.
8. Enter the following code.

```python
#!/usr/bin/env python3
"""
    Filename: bme680_simple_2.py
   Purpose: Use Pimoroni library to read
    temperature, pressure, and humidity from Bosch bme680 sensor
    !Connect to I2C bus
    Press Ctrl+C to exit
"""
from sys import exit
from time import sleep
# sudo pip3 install bme680
# Import bme680 library
import bme680

# Initialize sensor object, make connection to sensor over I2C
sensor = bme680.BME680(bme680.I2C_ADDR_PRIMARY)

print("BME680 Read temperature, pressure, and humidity")
print("Press CTRL+C to Exit")

try:
    while True:
        # Read the sensor
        if sensor.get_sensor_data():
            # If sensor data retrieval is successful,
            # retrieve and display the data

            # Temperature in celsius
            temp_c = sensor.data.temperature

            # Relative humidity in %
            humidity = sensor.data.humidity

            # Barometric pressure in hPa (hectopascal)
            pressure_hpa = sensor.data.pressure

            print(f"{temp_c:.1f} C | {humidity:.0f} % | {pressure_hpa:.2f} hPa")

            sleep(1)

except KeyboardInterrupt:
    print("Bye!")
    exit(0)
```

Example run:

```
pi@gpspi:~/Code $ python3 bme68x_read_tph_simple_1.py
Polling:
22.4 C  44 %  871.37 hPa
22.4 C  44 %  871.37 hPa
22.4 C  44 %  871.36 hPa
22.4 C  44 %  871.37 hPa
22.4 C  44 %  871.37 hPa
22.4 C  44 %  871.38 hPa
22.4 C  44 %  871.39 hPa
22.4 C  44 %  871.39 hPa
```

Tutorial 3: BME680 Simple 3

Barometric pressure compensation for altitude:
https://www.engineeringtoolbox.com/barometers-elevation-compensation-d_1812.html

```python
#!/usr/bin/env python3
"""
    Filename: bme680_simple_3.py
    Description: Display temperature, pressure, and humidity
    from Bosch bme680 sensor
    !Connect to I2C bus
    Press Ctrl+C to exit
"""
from sys import exit
from time import sleep
# sudo pip3 install bme680
# Import bme680 library
import bme680

# You will usually have to add an offset to account for the temperature of
# the sensor. This is usually around 5 degrees but varies by use. Use a
# separate temperature sensor to calibrate this one.
TEMP_OFFSET = -1

# Initialize sensor object, make connection to sensor over I2C
sensor = bme680.BME680(bme680.I2C_ADDR_PRIMARY)

print("Press CTRL+C to Exit")
print("Polling:")
```

```python
try:
    while True:
        # Check if sensor data can be retrieved successfully.
        if sensor.get_sensor_data():
            # Sensor output in celsius
            temp_c = sensor.data.temperature
            # Convert celsius to fahrenheit
            temp_f = ((temp_c * 9.0) / 5.0) + 32
            # Correct temperature
            temp_comp = temp_f + TEMP_OFFSET

            # Measure relative humidity in %
            humidity = sensor.data.humidity

            # Sensor output in hectoPascals (hPa), also called millibars
            pressure_pascals = sensor.data.pressure
            # Convert hPa hectopascals to inHg Inches of Mercury
            pressure_inhg = pressure_pascals / 33.863886666667
            # Compensate for 3960' altitude 4.04
            # Scottsbluff, NE, Heilig Field, 4.04
            pressure_inhg = pressure_inhg + 4.04

            print()
            print(f" Temp comp: {temp_comp:.1f} °F")
            print(f"  Humidity: {humidity:.1f}%")
            print(f"  Pressure: {pressure_inhg:.2f} inHg")
            print(f"  Pressure: {pressure_pascals:.2f} KPa")

            sleep(5)


except KeyboardInterrupt:
    print("Bye!")
    exit(0)
```

Example run:

```
Press CTRL+C to Exit
Polling:
69.2 °F - 42.4% - 29.83 inHg - 873.37 KPa
69.2 °F - 42.4% - 29.83 inHg - 873.38 KPa
69.2 °F - 42.5% - 29.83 inHg - 873.38 KPa
69.2 °F - 42.5% - 29.83 inHg - 873.39 KPa
```

Air Quality

The sensor needs to be calibrated for 5 days when it is first used. The sensor value is not accurate before/during calibration. Wait about 2 mins after module powered on for gas heater enter a steady state. The gas value at this time is correct.

The IAQ value corresponding air quality:

0-50: Good

51-100: Moderate

101-150: Unhealthy for Sensitive Groups

151-200: Unhealthy

201-300: Very Unhealthy

301-: Hazardous

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bme68x_test.py

This program will read the BME680 every 5 seconds and display to the console.